

YEAR 7 · UNIT 2 · HOMEWORK 6

Letters and Particles

Mathematics 2.1–2.2 — expressions, formulae and substituting

Science 2.1–2.2 — states of matter, particles, changes of state

Section C — the one thing both subjects measured this week

Name: _____

Class: _____

Date given: _____

Date due: **Monday 14 September 2026****How to do this homework**Three sections. Maths first, then science, then **Section C**, where the two meet.

- In maths, **write the middle line** — put the times signs back in first. And remember that $c - 50$ is a **finished answer**: you are allowed to stop.
- In science, answer in **full English sentences**. “Condensation” is not a sentence; “This is condensation, from gas to liquid” is.

Section A · Mathematics 2.1–2.2 — Expressions and Formulae**A1 Expression, formula, or neither?**

Tick one box in each row. The first one is done for you.

The two definitions, side by side

An **expression** contains letters and sometimes numbers, and has **no equals sign**. For example: $n + 7$, $4m$, $x - 3$.

A **formula** is a rule connecting two or more quantities. It contains letters and **has an equals sign**. For example: $A = lw$, $y = 5x$.

	Expression	Formula	Neither
$6h$	X		
$P = 4s$			
$8 + 5 = 13$			
$t - 9$			
$V = lwh$			
$m = 4 + n$			

A2 Write the expression

Mr Bowen has a beaker holding w grams of salt. Write an expression for the mass of salt in each of these.

Word help – the phrases, and the two that decide the operation

more than · fewer than	add · subtract
times as many / as much	multiply
half as much	divide by 2
the total of	add them together
the difference between	subtract the smaller from the larger

Write $3 \times s$ as **3s**, and $s \div 2$ as **s over 2**. The number always goes **in front** of the letter – never $s3$.

- (a) 30 grams more than Mr Bowen _____
- (b) Four times as much as Mr Bowen _____
- (c) 25 grams fewer than Mr Bowen _____
- (d) Half as much as Mr Bowen _____

Now two beakers. The first holds p grams and the second holds q grams. Write an expression for:

- (e) the **total** mass of salt in the two beakers _____
- (f) the **difference** between the two masses _____
- (g) the mass in **six** beakers like the first and **two** like the second _____

A3 The word that changes the order**Word help – read these twice before you start**

subtract 6	you take 6 away from what you have: $4x - 6$
subtract from 6	you start at 6 and take away what you have: $6 - 4x$
h less than t	you start at t : $t - h$. The English gives the two letters in the opposite order to the calculation.
the result	the thing you just worked out on the line before

Write an expression for each description.

- (a) Multiply x by 7, then subtract 3. _____
- (b) Multiply x by 7, then subtract the result from 3. _____

(c) g less than k _____(d) y more than nine times z _____(e) Multiply n by 4, then subtract the result from 30. _____

(f) Mr Bowen looks at the expression $9 - 2m$ and says: "Multiply m by 2, then subtract 9."

Is he right? If he is wrong, write the correct description.

A4 Substitute — and put the times sign back in

The middle line

$4n$ was only ever a short way of writing $4 \times n$. So when $n = 3$:

$$4n = 4 \times 3 = 12 \quad \text{not } 43.$$

The middle line is where the marks are. **Write it every time.**

Work out the value of each expression when $n = 6$. Write the middle line.

(a) $n + 8 =$ _____

(c) $n - 11 =$ _____

(b) $5n =$ _____

(d) $\frac{n}{2} =$ _____

A5 The order of operations does not switch off

Times and divide happen first

Even after you have put a number in place of a letter, **times and divide** still happen **before** add and subtract.

$$3x + 2 \quad \text{when } x = 4 \quad \text{is} \quad 3 \times 4 + 2 = 12 + 2 = 14, \quad \text{not } 18.$$

Work these out. **Show the middle line** — an answer with no working cannot be checked.

(a) $6x - 5$ when $x = 4$ _____

(b) $30 - 4n$ when $n = 7$ _____

(c) $2a + 3b$ when $a = 5$ and $b = 4$ _____

(d) $\frac{k}{3} + 9$ when $k = 24$ _____

A6 The minus sign travels with its number**Use brackets**

When the number you are substituting is **negative**, put it in **brackets** so its minus sign cannot be left behind.

$$4 - 3n \text{ when } n = -2: \quad 3n = 3 \times (-2) = -6, \text{ so } 4 - (-6) = 4 + 6 = 10.$$

Work these out when $m = -5$.

(a) $2m =$ _____

(c) $1 - 2m =$ _____

(b) $8 - m =$ _____

A7 Use a formula

Mr Bowen heats a beaker of water. He finds that the temperature in degrees Celsius is given by the formula

$$T = 22 + 6t, \text{ where } t \text{ is the number of minutes since he lit the burner.}$$

(a) Find T when $t = 5$. _____

(b) Find T when $t = 11$. _____

(c) Mr Bowen puts $t = 40$ into the formula and gets $T = 262$. His arithmetic is correct. Explain, using what you know from **Science 2.2**, why the water will never reach that temperature.

A8 Now you write the question**Read all four conditions before you start**

Here is an expression:

$$40 - 5n$$

Write a **word problem** that this expression is the answer to. Your question must:

1. be about something **real** — a beaker, money, a journey, a class;
2. say clearly what **n represents**;
3. **start** at 40, and take away 5 lots of something;
4. be written in **full sentences** and **end in a question**.

Section B · Science 2.1–2.2 – Particles and Changes of State

B1 Match the word to its meaning

Write the correct **letter** in each box.

Word		Meaning
1. Matter		A A space with no particles in it at all.
2. Property		B Squashed into a smaller space.
3. Particle		C The amount of space something takes up.
4. Volume		D Everything you can see and feel.
5. Compressed		E A tiny piece of matter, far too small to see.
6. Vibrate		F A way that a substance behaves.
7. Vacuum		G The curved surface of a liquid in a tube.
8. Meniscus		H To make tiny movements without leaving your place.

B2 The properties table

Put a **tick** for yes and a **cross** for no.

	Solid	Liquid	Gas
Keeps its own shape	☐	☐	☐
Keeps the same volume	☐	☐	☐
Can be poured	☐	☐	☐
Can be compressed	☐	☐	☐

(b) Dry sand can be poured from one hand to the other. Sand is **not** a liquid. Explain why not.

B3 Answer in full sentences**Sentence starters – you may copy these**

“In a solid the particles are...”

“This means that...”

“...because the gaps between the particles...”

(a) You can pour water, but you cannot pour a brick. Use **particles** to explain the difference.

(b) Mr Bowen blocks the end of two syringes with his thumb. One is full of **air** and one is full of **water**. Only the air syringe moves when he pushes. Explain why.

(c) A kitchen sponge is a solid, but you can squash it in one hand. Does this prove that particle theory is **wrong**? Explain your answer.

B4 The five changes of state

Complete the table. The first row is done for you.

Doing word	Naming word	From ... to ...	Heat or cool?
melt	melting	solid to liquid	heat
freeze			
	evaporation		
boil			
	condensation		

(b) Write the two temperatures for water:

melting point _____ boiling point _____

B5 Evaporating or boiling?**Same journey, different word**

Evaporation and **boiling** are both liquid to gas. Evaporation is **slow**, happens at the **surface** only, and at **any** temperature. Boiling is **fast**, happens **all through** the liquid, and only at the **boiling point**.

Tick one box in each row.

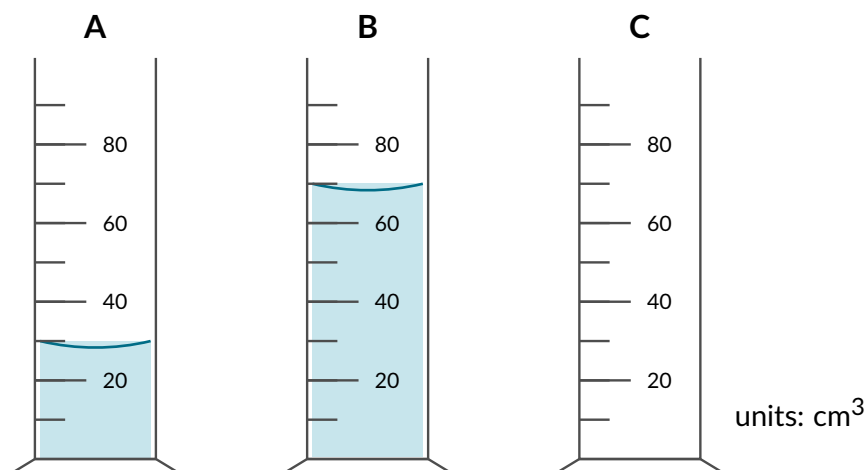
	Evaporating	Boiling
Washing dries on the line on a cool day		
Bubbles form all through the water in a pan		
A puddle disappears from the playground		
It only happens at 100 °C		

(b) Mr Bowen leaves a glass of water on his desk on Friday. On Monday the glass is empty. The water never reached 100 °C. Name the change of state and explain, in full sentences, where the water went.

B6 Read the scale, then draw one**Both instruments, one rule**

Read at **eye level**. On a measuring cylinder, read the **bottom of the meniscus**. On a thermometer, read the **top** of the liquid.

Each small line on the cylinders is 10 cm³. Write the volume in **A** and **B**, then **draw** the water level in **C**.



A = _____

B = _____

C: draw 45 cm³

(b) Two students read cylinder **A** and get two different answers. Neither of them misread the numbers. Explain what went wrong.

Section C · Where the two subjects meet – Volume

In Science you measured a volume in a cylinder. In Maths you built one out of letters. They are the same thing, and this section is why.

The small number counts the lengths

A **length** is one measurement: cm.

An **area** is a length **times** a length: $\text{cm} \times \text{cm} = \text{cm}^2$. Two lengths, so a small **2**.

A **volume** is a length **times** a length **times** a length: $\text{cm} \times \text{cm} \times \text{cm} = \text{cm}^3$. Three lengths, so a small **3**.

So for a box: $V = lwh$ — three letters multiplied, three lengths, cm^3 .

And the unit that joins the two subjects: $1 \text{ cm}^3 = 1 \text{ ml}$, $1000 \text{ cm}^3 = 1 \text{ litre}$.

C1 Count the letters

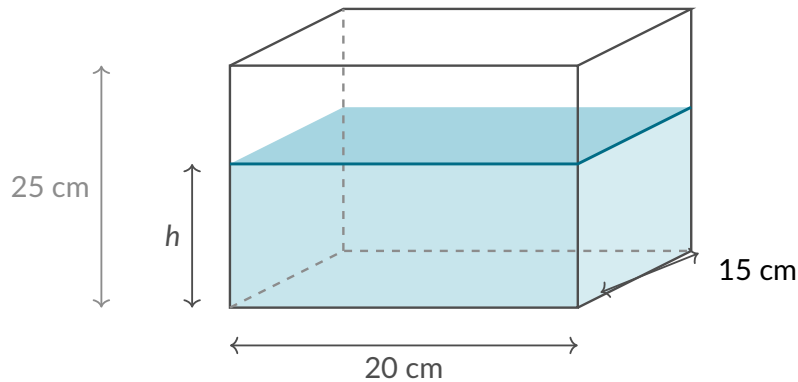
Every letter below stands for a length in centimetres. Count how many lengths are **multiplied together**, then write the unit. The first one is done for you.

Expression	How many lengths multiplied?	Unit
h	one	cm
lw		
lwh		
s^3		
$2l + 2w$		

(b) The last row is different from all the others. Explain why the unit for $2l + 2w$ is **not** cm^2 , even though it has two letters in it.

C2 The tank

A tank has a rectangular base measuring **20 cm** by **15 cm**. It is **25 cm** tall. Mr Bowen pours in water to a depth of h centimetres.



- Write an **expression** for the volume of the water in the tank. _____
- Write it as a **formula** beginning $V =$ _____
- Find the volume of water when $h = 4$. Show the middle line. _____
- How many **litres** of water is that? _____
- Mr Bowen substitutes $h = 40$ into his formula and gets $12\,000\text{ cm}^3$. His arithmetic is correct. Explain why the answer is still wrong.

C3 The same water, three containers

Mr Bowen pours the **same** 150 cm^3 of water from a tall narrow cylinder into a wide flat dish, and then into a beaker.

- As the water moves from container to container, which of the three measurements – its **length**, its **width**, its **height** – change? _____
- What is the value of lwh in every one of the three containers? _____
- Complete the science sentence:
A liquid keeps the same _____ but takes the _____ of its container.
- So l , w and h all change, and lwh does not. Explain in your own words how those two facts fit together.

C4 The syringe — where the volume really does change

A sealed syringe holds $v \text{ cm}^3$ of **air**. Nothing can get in or out.

- (a) Mr Bowen pushes the plunger until the air fills **half** the space it did before. Write an expression for the new volume of the air. _____
- (b) Mr Bowen empties the syringe, fills it with $v \text{ cm}^3$ of **water** instead, seals it, and pushes just as hard. Write an expression for the new volume of the water, and say why.

C5 Three bad ideas

Mr Bowen has had three ideas. Each one is a bad idea. For each, say **what goes wrong**, in a full sentence, using the science or the maths from this unit.

1. He wants a smaller cup of tea, so he seals the cup and squeezes it very hard.
- _____
- _____
2. His formula says the water will be at 262°C after 40 minutes, so he plans to leave the burner on and come back at lunchtime.
- _____
- _____
3. He measures a box as 10 cm long, 10 cm wide and 10 cm tall, and writes that its volume is 30 cm^3 .
- _____
- _____
- _____